

EE-312L Signals and Systems Lab
Laboratory Session 05
**Understanding The Discrete-Time Sequences
and Their Transformation Using MATLAB**

October 18, 2024

Outlines:

1. Objective
2. Unit Step Function
3. Unit Rect Function
4. Unit Ramp Function
5. Complex Exponential Sequence

1 Objective

The objective of this lab session is to understand discrete-time sequences and their transformation using MATLAB.

2 Discrete-Time Signals

2.1 Signum Function

$$\text{sgn}[n] = \begin{cases} 1, & n > 0 \\ 0, & n = 0 \\ -1, & n < 0 \end{cases}$$

The **sign(n)** function in MATLAB can be used to generate the plot of signum function.

2.2 Unit-Step Function

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

2.3 Unit-Ramp Function

$$ramp[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

2.4 Rectangular Function

$$rect[n] = \begin{cases} 1, & |n| \leq N \\ 0, & |n| > N \end{cases}$$

2.5 Triangular Function

$$tri[n] = \begin{cases} 1 - 2|n|, & |n| \leq N \\ 0, & \text{Otherwise} \end{cases}$$

2.6 Discrete-Time Complex Exponential Sequence

$$\begin{aligned} x[n] &= A\alpha^n \\ &= |A||\alpha|^n e^{j(\omega_0 n + \theta)} \\ &= |A||\alpha|^n [\cos(\omega_0 n + \theta) + j\sin(\omega_0 n + \theta)] \end{aligned}$$

where

- $A = |A|e^{j\theta}$, a complex number
- $\alpha = |\alpha|e^{j\omega_0}$, a complex number
- ω_0 , discrete frequency in radians

3 Generating Discrete-Time Sequences Using MATLAB

We will be using the following functions:

3.1 Function for generating unit step function

```
1 function y = ustepdt(n, ad)
2     % n: sequence number
3     % ad: advance (for +ve value) delay(for -ve value),
4     L = length(n);
5     y = zeros(1,L);
6     for i = 1:L,
7         if n(i) >= -ad,
8             y(i) = 1;
9         endif
10    endfor
```

3.2 Function for generating unit ramp function

```
1 function y = ramp(n, m, ad)
2     % n: sequence number
3     % m: slope
4     % ad: advance (for +ve value) delay(for -ve value),
5     N = length(n);
6     y = zeros(1,N);
7     for i = 1:N,
8         if n(i) >= -ad,
9             y(i) = m*(n(i) + ad);
10        endif
11    endfor
```

3.3 Function for generating rect function

```
1 function y = rectdt(n, N, nd)
2     % n: sequence number
3     % 2N+1: width of the rect function
4     % ad: advance (for +ve value) delay(for -ve value),
5     y = ustepdt(n,N+nd) - ustepdt(n,-N+nd);
```

3.4 MATLAB code for generating plots

```
1
2 clear all, clc, close all;
3 n = -7:1:7;
4
5 %%% Signum Function %%%
```

```
6
7 figure(1)
8 y = sign(n);
9 stem(n,y, 'LineWidth',2.5)
10 xlabel('n');
11 ylabel('sgn[n]');
12 title('Signum function plot using MATLAB');
13 ylim([-2 2])
14 xlim([-7 7])
15 grid minor on
16 saveas(gcf, 'lab05plots/plot01.pdf')
17
18 %%% Unit step %%%
19
20 figure(2)
21 stem(n,ustepdt(n,0), 'r', 'linewidth',2.5);
22 title('Unit step signal u[n]')
23 axis([-7 7 -3 3]);
24 legend('Unit Step');
25 xlabel('n');
26 ylabel('u[n]');
27 grid minor on;
28 saveas(gcf, 'lab05plots/plot02.pdf')
29
30 figure(3)
31
32 subplot(2,1,1)
33 stem(n,ustepdt(n,3), 'r', 'linewidth',2.5);
34 axis([-7 7 -3 5]);
35 legend('Unit Step advanced by 3 units');
36 xlabel('n');
37 ylabel('u[n+3]');
38 grid minor on;
39
40 subplot(2,1,2)
41 stem(n,ustepdt(n,-3), 'r', 'linewidth',2.5);
42 axis([-7 7 -3 5]);
43 legend('Unit Step delayed by 3 units');
44 xlabel('n');
45 ylabel('u[n-3]');
46 grid minor on;
47 saveas(gcf, 'lab05plots/plot03.pdf')
48
49
50 %%% Unit Rect Function %%%
```

```
51 N = 3;
52 figure(4)
53
54 subplot(3,1,1)
55 stem(n,rectdt(n,N,0),'r','linewidth',2.5);
56 axis([-7 7 -3 3]);
57 legend('Rect function');
58 xlabel('n');
59 ylabel('rect[n]');
60 grid minor on;
61
62 subplot(3,1,2)
63 stem(n,rectdt(n,N,2),'r','linewidth',2.5);
64 axis([-7 7 -3 3]);
65 legend('Rect function');
66 xlabel('n');
67 ylabel('rect[n+2]');
68 grid minor on;
69
70 subplot(3,1,3)
71 stem(n,rectdt(n,N,-2),'r','linewidth',2.5);
72 axis([-7 7 -3 3]);
73 legend('Rect function');
74 xlabel('n');
75 ylabel('rect[n-2]');
76 grid minor on;
77 saveas(gcf,'lab05plots/plot04.pdf')
78
79 %%% Unit Ramp %%%
80
81 figure(5)
82
83 subplot(2,1,1)
84 stem(n,rampdt(n,1,0),'r','linewidth',2.5);
85 axis([-7 7 -3 7]);
86 legend('Ramp function');
87 xlabel('n');
88 ylabel('ramp[n]');
89 grid minor on;
90
91 subplot(2,1,2)
92 stem(n,rampdt(-n,1,2),'r','linewidth',2.5);
93 axis([-7 7 -3 7]);
94 legend('Ramp with time reversal');
95 xlabel('n');
```

```
96 ylabel('ramp[-n]');
97 grid minor on;
98 saveas(gcf,'lab05plots/plot05.pdf')
99
100
101 figure(6)
102
103 subplot(2,1,1)
104 stem(n,rampdt(n,1,0),'r','linewidth',2.5);
105 axis([-7 7 -3 7]);
106 legend('Ramp function');
107 xlabel('n');
108 ylabel('ramp[n]');
109 grid minor on;
110
111 subplot(2,1,2)
112 stem(n,rampdt(n,1,3),'r','linewidth',2.5);
113 axis([-7 7 -3 7]);
114 legend('Ramp with time shift');
115 xlabel('n');
116 ylabel('ramp[n+3]');
117 grid minor on;
118 saveas(gcf,'lab05plots/plot06.pdf')
119
120
121 %saveas(gcf,'plots/filename.pdf')
```

3.5 MATLAB Generated plots for $u[n]$, $rect[n]$ and their time-shift, and time-reversal

3.6 MATLAB Generated plots for Real Exponential Signals

1. $x_1[n] = 0.8^n$, exponentially decaying signal
2. $x_2[n] = 1.25^n$, exponentially growing signal
3. $\gamma_1 = x_1[n] \cos(\pi n)$, modulated exponential
4. $\gamma_2 = x_2[n] \cos(\pi n)$, modulated exponential

```
1
2 clear all, clc, close all;
3 n = -7:1:7;
4
5 %%% Discrete-Time Real Exponential Signals %%%
```

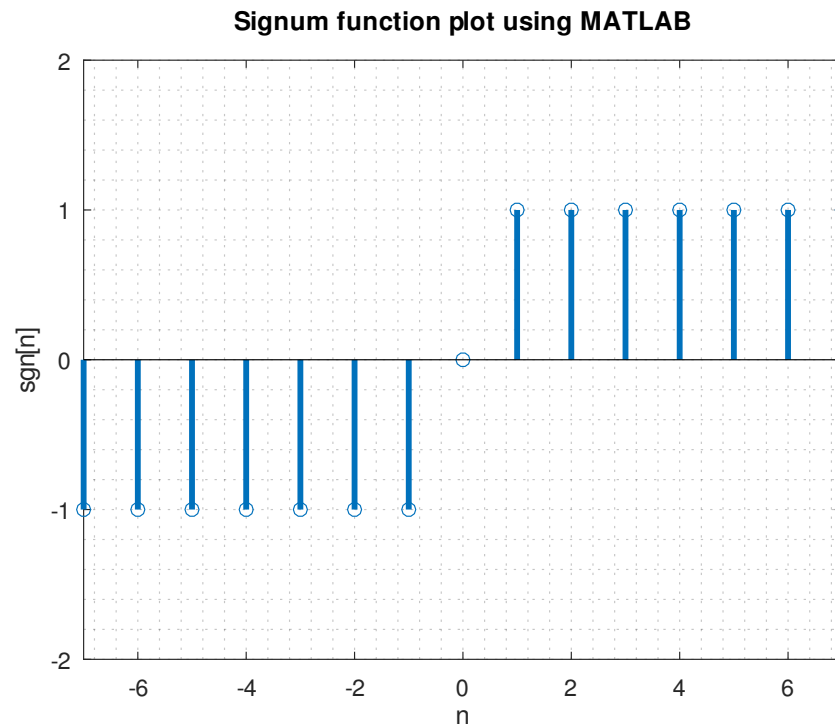


Figure 1: Discrete-time signum function

```
6
7 x1 = 0.8.^n;
8 x2 = 1.25.^n;
9 gamma1 = x1.*cos(pi*n);
10 gamma2 = x2.*cos(pi*n);
11
12
13 figure(1)
14 subplot(2,2,1)
15 y = sign(n);
16 stem(n,x1, 'LineWidth',2.5)
17 xlabel('n');
18 ylabel('x_1[n]');
19 title('Decaying Exponential');
20 ylim([-1 4])
21 xlim([-6 6])
22 grid minor on
23
24 subplot(2,2,2)
25 y = sign(n);
26 stem(n,x1, 'LineWidth',2.5)
27 xlabel('n');
```

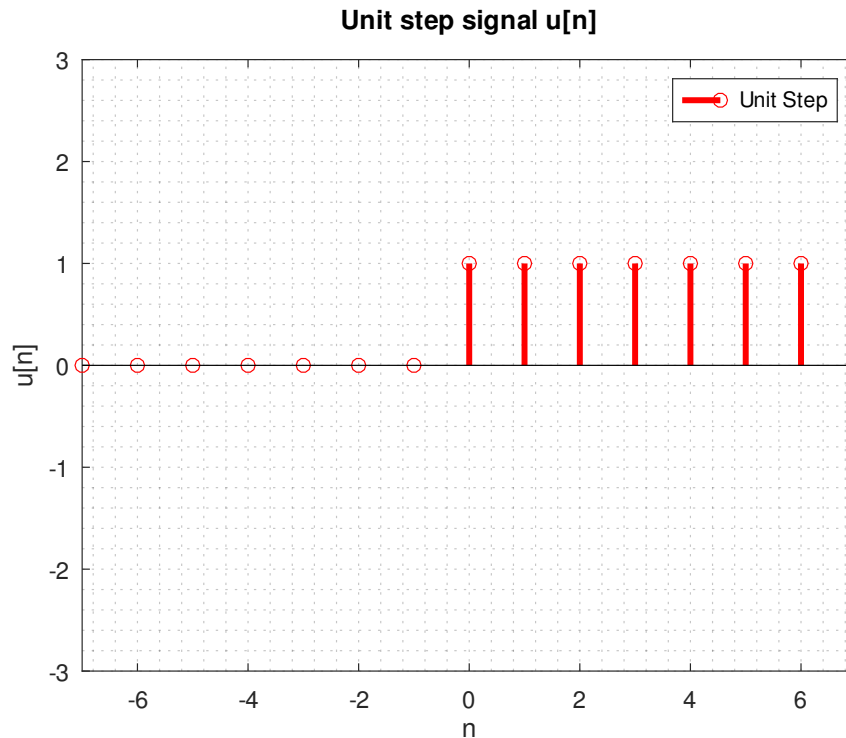


Figure 2: Discrete-time Unit step function

```

28 ylabel('x_1[n]');
29 title('Growing Exponential');
30 ylim([-1 4])
31 xlim([-6 6])
32 grid minor on
33
34 subplot(2,2,3)
35 y = sign(n);
36 stem(n,gamma1, 'LineWidth',2.5)
37 xlabel('n');
38 ylabel('\gamma_1[n]');
39 title('Modulated Exponential');
40 ylim([-4 4])
41 xlim([-6 6])
42 grid minor on
43
44 subplot(2,2,4)
45 y = sign(n);
46 stem(n,gamma1, 'LineWidth',2.5)
47 xlabel('n');
48 ylabel('\gamma_2[n]');
49 title('Modulated Exponential');

```

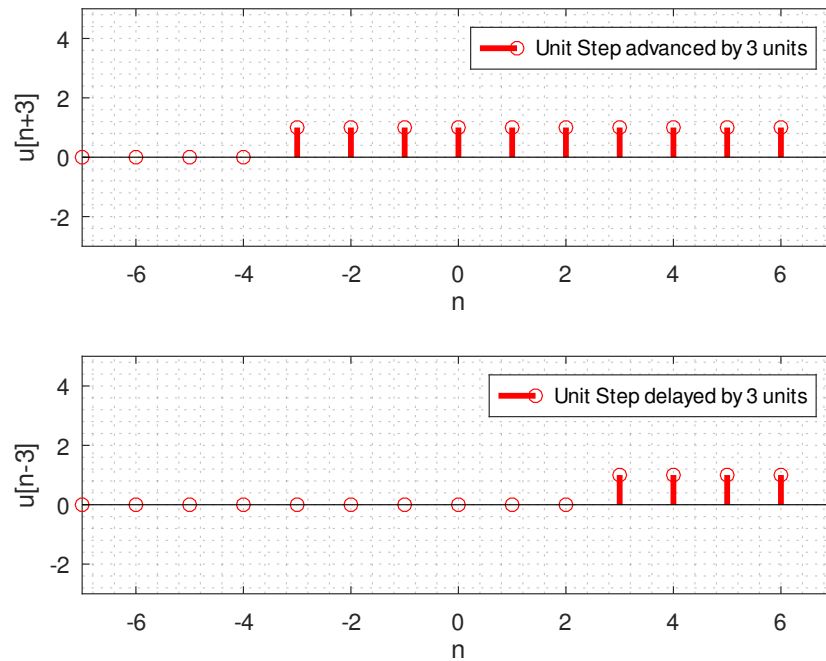



Figure 3: Discrete-time Unit step function with time shift

```
50 ylim([-4 4])
51 xlim([-6 6])
52 grid minor on
53
54
55 saveas(gcf, 'lab05plots/plot07.pdf')
```

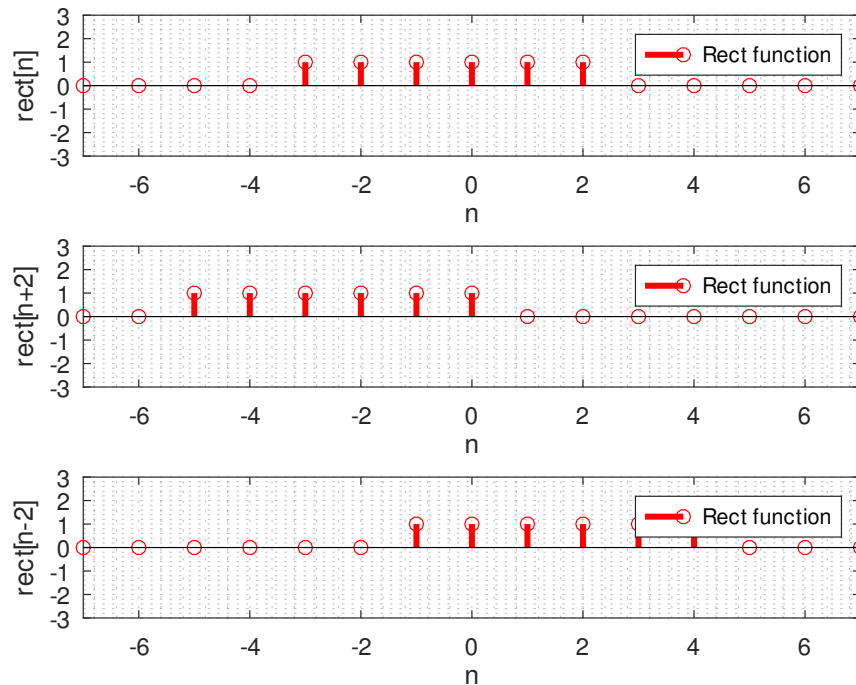


Figure 4: Discrete-time unit rect function

4 Lab 05: Lab Tasks

1. Consider the discrete-time sequence $\gamma[n]$,

$$\gamma[n] = \begin{cases} 0, & n \leq -21 \text{ or } n \geq 21 \\ 0.45n + 9, & -20 \leq n \leq -7 \\ -0.45n + 3, & -6 \leq n \leq 0 \\ 3, & 1 \leq n \leq 19 \\ 0, & n > 20 \end{cases} \rightarrow -27 \leq n \leq -21$$

Use MATLAB to plot

- (a) $\gamma[n]$
- (b) Even component of $\gamma[n]$.
- (c) Odd component of $\gamma[n]$.

2. Consider the following discrete-time sequence

$$x[k] = \begin{cases} k & 0 \leq k \leq 3, \\ 0 & \text{otherwise} \end{cases}$$

Use MATLAB to plot

- (a) $x[k]$
- (b) $x[-k]$

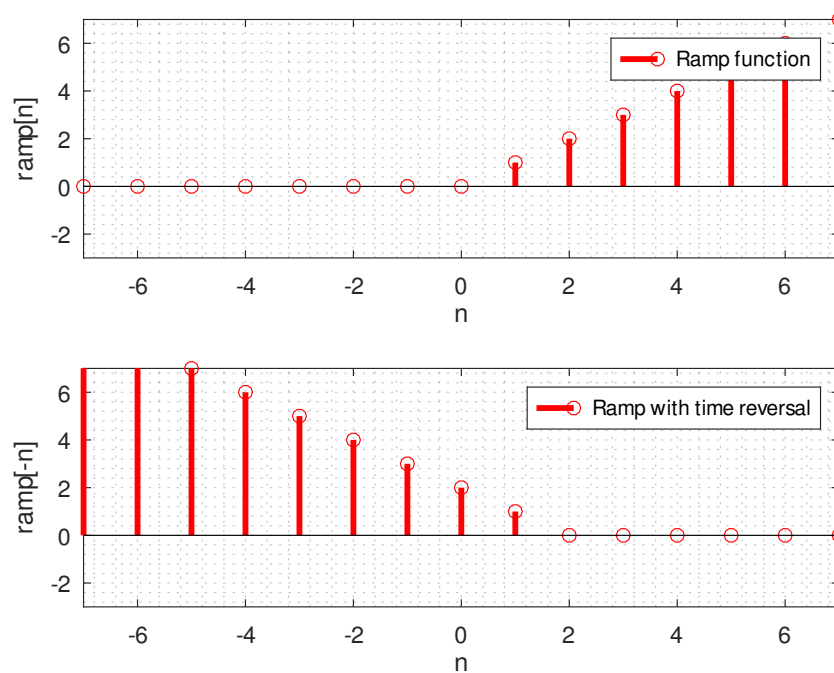


Figure 5: Discrete-time unit ramp function with time scaling

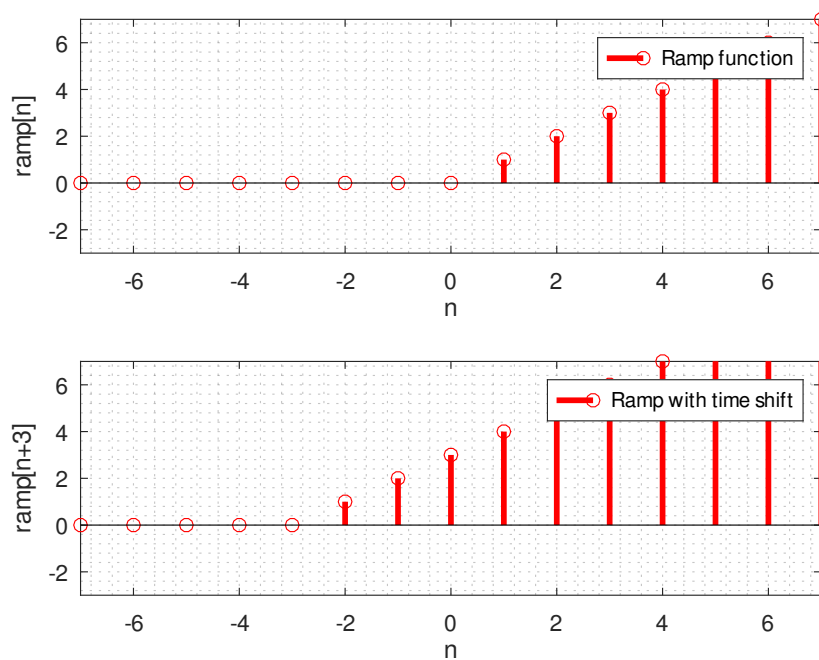


Figure 6: Discrete-time unit ramp function with time shift

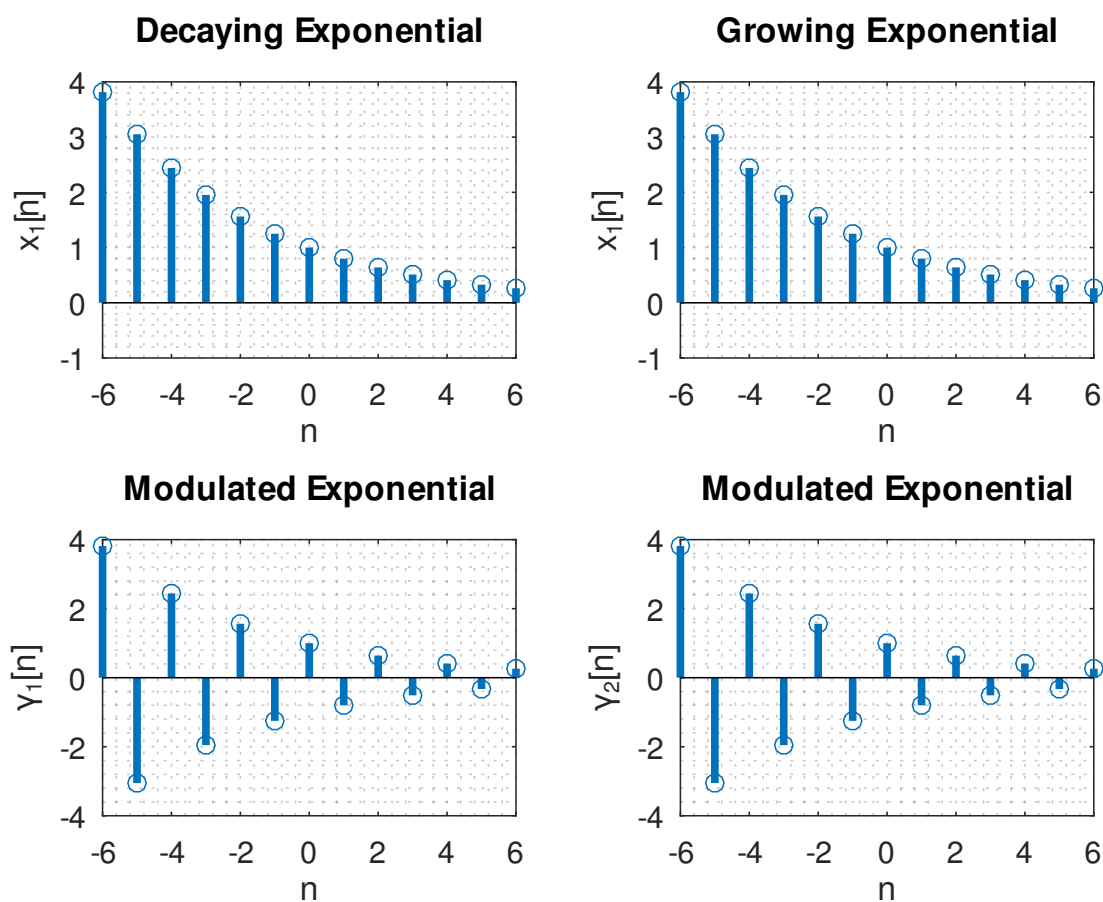


Figure 7: Real exponential functions